



**BLANTYRE INTERNATIONAL  
UNIVERSITY**

# **Bachelors Degree in Information Technology**

## **VISUAL PROGRAMMING ASSIGNMENTS (IT 212)**

(JAN–JUNE, 2020)

**NB: Please indicate your name on the assignment**

**INSTRUCTIONS:**

1. Both assignments should be submitted in form of zip files on **whatsapp** with the name indicated or directly to the lecturer.
2. Please confirm that your solution is working before submission.
3. **Any copied work will attract a zero for both students**
4. Late submission will not be tolerated (assignment won't be marked)
  - a. Due date for first assignment: **25<sup>th</sup> March , 2020**
  - b. Due date for first assignment: **5<sup>th</sup> May , 2020**

### Assignment1 (50 marks)

1. Create a guess game that allows a user to guess and enter a number from 1 to 10. Then the program will also generate its number and compare with the number entered by the user. If numbers are equal, the program should display **you won!** on a message box having an information icon. Otherwise, the program should display the message that they lost! The number I guessed is 5. On a message box. If they lose, the program should state whether the number is too high or too low to the number the program generated.

For example: you lost! Number too high, I guessed 6

The user should be allowed to play for 1 minute. Then the program should display a message box: **game over!** And exit the program

2. Write a program that calculates a monthly payroll for an employee and prints the details to a file. The program should accept monthly gross salary and produce the payslip. The program should use the following information to produce the payslip

**PAYE 10% of gross**

**Medical scheme 2% of gross**

**Pension 3% of gross**

After the deductions, the program should print all the details such as name of employee, gross salary, basic pay, pension, medical contribution to a file named **payslip.txt**

### Assignment2 (50 marks) please refer to chapter15 in the module

Develop an employee management system that will help to keep and manage employee details.

The details are as follows:

The first page should present the admin with a menu as shown below

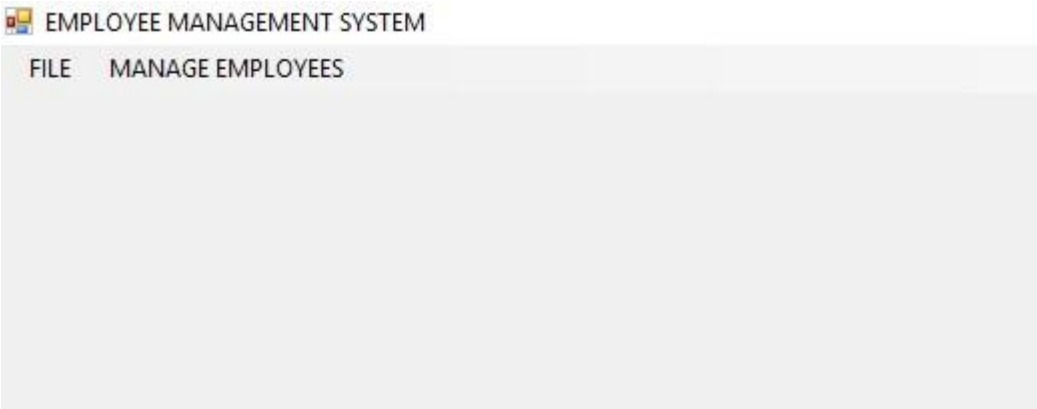


FIGURE 1: First Form

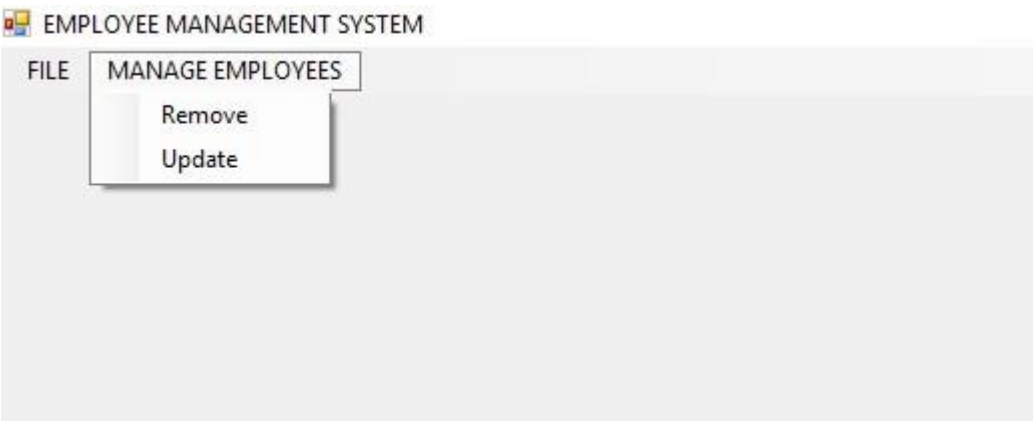


FIGURE 1: Manage employees menu

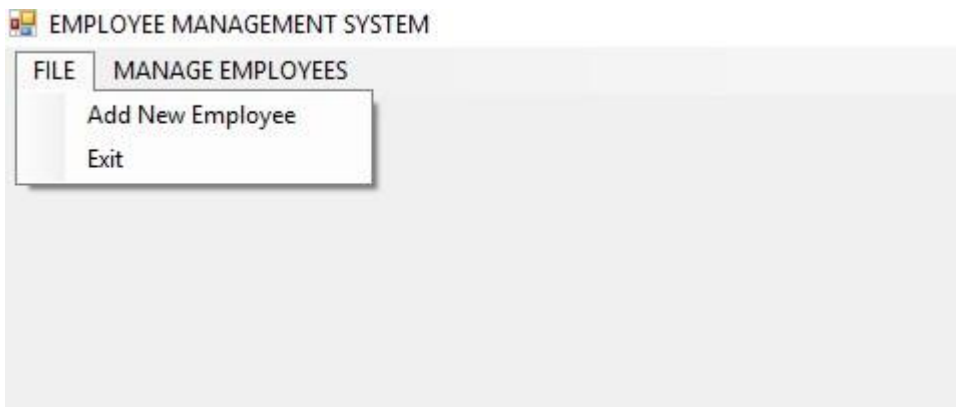


FIGURE 2: File menu

The admin should be able to **Remove, update** and **Add employees** to the system( the add new form should have text boxes corresponding to those shown on the data

grid in **figure 4** below except for the ID which should be an auto number hence should not be entered.

The **exit** menu option should close the system. Refer to the pictures below.

Note: each form i.e. delete , add new or update should depict the same appearance with a button to perform the task i.e. on delete you should have a delete button to allow the admin to delete a selected record from the data grid and the same applies to the rest of the forms. They should have a corresponding button to perform the task supposed at that form. If the admin tries to delete or update a record before selecting the record the system should produce a message box telling them to select the record on the data grid first.



|   | ID | FIRSTNAME | LASTNAME | POSITION   | SALARY |
|---|----|-----------|----------|------------|--------|
| ▶ | 5  | Dan       | Liwonde  | Manager    | 20000  |
|   | 6  | John      | Kawale   | Director   | 10000  |
|   | 7  | Grace     | Ngale    | Supervisor | 70000  |
| * |    |           |          |            |        |

FIGURE 4 :Delete employee form